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Contagion risk prediction with Chart Graph Convolutional Network: Evidence from Chinese stock market

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ABSTRACT

Contagion risk has garnered widespread attention with the major occurrence of financial crises. However, existing literature has mainly focused on the characteristics analysis of risk contagion, while ignoring the prediction of contagion risk. Therefore, considering the importance of time series charts, a framework for contagion risk prediction is proposed by comprehensively mining the features of technical charts and technical indicators. In detail, correlation and centrality of the financial network are extracted to specify the contagion risk. Then, time series charts are converted to graphs by the chart similarity, and each node in the graph is assigned with technical indicators. Finally, a Chart Graph Convolutional Network (Chart GCN) is used to extract the features in the graph and predict the contagion risk. Based on the 28 sectors of the Chinese stock market, we specify the contagion risk and verify its consistency with the occurrence of financial crisis. In addition, our model proves the effectiveness of mining time series chart features in stock market contagion risk prediction. These results are essential for risk management in the financial market.

1. Introduction

The financial market has been consistently impacted by financial crises and significant epidemic events (Chen et al., 2022; Zhao et al., 2022). These events have led to huge economic losses and further highlighted the growing contagion risk (Xu et al., 2022). Specifically, contagion risk refers to the spread of economic recession from one market or sector to another (Pais and Stork, 2011; Daly et al., 2019; Gong et al., 2020). A large body of literature has mostly attempted to analyze the characteristics of financial risk contagion from multiple perspectives of time scale (Wen et al., 2022; Xu et al., 2022) and complex network (Chen et al., 2020; Li et al., 2021b; Abduraimova, 2022), and less attention has been paid on the specification and prediction of contagion risk. In addition, the Chinese financial system has developed rapidly and the stock market is an important part of Chinese financial system (Wu et al., 2019; Chen et al., 2020). Despite its second largest scale around the world, the Chinese stock market has often been considered as a contagion risk arena. Different sectors in the stock market have become closely interconnected with risk transmitting rapidly through contagion mechanisms (Pan and Mishra, 2018; Li et al., 2021b). Therefore, it is essential to specify and predict the contagion risk among different sectors in Chinese stock market.

Existing scholars have studied the risk contagion characteristics of financial markets from multiple perspectives. Chen et al. (2020) analyzed how risk contagion varies across banking network structures and identified the most important and vulnerable nodes. Li et al. (2021b) captured the risk contagion effect of different regions in Chinese stock market based on inter-regional volatility spillover network. Abduraimova (2022) proposed a new contagion measure to detect the tail risk contagion of international

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stock markets. Wen et al. (2022) examined the dynamic dependence and risk contagion effects under multiple time scales among oil market, commodity market and stock market. Xu et al. (2022) accounted for the interdependency of tail risks among sectors in Chinese stock market and identified risk contagion through systematic risk decomposition techniques.

Recently, studies have begun to measure and specify contagion risk in financial market. Daly et al. (2019) proposed extreme value theory to measure multiple contagion risk within and across the Australian banking sector. Gong et al. (2020) calculated a downward dependency factor to assess the contagion risk. To the best of our knowledge, only one paper is found to clearly specify the definition of contagion risk (Samitas et al., 2020). The paper specifies the initial occurrence time of contagion risk in the international financial markets by capturing the time node of the correlation increasing sharply while the centrality growing intensely. If two sectors exhibit a high level of correlation during tranquil periods and continue to be highly correlated after one of them faces a shock, it may not constitute financial contagion. It is noteworthy that they utilized the asymmetric dynamic conditional correlation (ADCC)-GJR-GARCH model, which takes into account the asymmetry of contagion risk, to construct their complex network. They applied a machine learning algorithm named Multilayer Perceptron (MLP) to predict the contagion risk.

However, deep learning methods have been mostly employed to enhance the accuracy of risk prediction in the recent studies. To be specific, deep learning models include Deep Neural Networks (DNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), and PATCHY-SAN (Select-Assemble-Normalize) (Niepert et al., 2016; Nelson et al., 2017; Zhao et al., 2017; Li et al., 2021a). Scholars preferred technical indicators over technical charts, although the effectiveness of time series charts has already been proven in stock movement prediction (Chatzis et al., 2018; Batola, 2019; Tölö, 2020; Li et al., 2022; Hanauer and Kalsbach, 2023; Hao et al., 2023). Compared to traditional deep learning models, Chart Graph Convolutional Network (Chart GCN) offers distinct advantages in terms of graph similarity, which can aid in better analysis and understanding of the structures, relationships, and patterns within time series graph data (Niepert et al., 2016; Xu et al., 2019; Chen et al., 2021; Li et al., 2022). It takes graph structure and node features as inputs, aggregates and transforms information from neighbors of each node to create a new representation for that node (Jafari and Haratizadeh, 2022). This representation, which is more informative for predicting target variables, can discover the hidden relationship between stock time series from the graph structure (Miao et al., 2023; He et al., 2024). The way that Chart GCN handles graphs is analogous to how CNNs process images, with CNNs demonstrating strong performance in extracting local image features (Niepert et al., 2016). However, to the best of our knowledge, the Chart GCN model has been rarely utilized in predicting the contagion risk of financial markets.

In addition, Chinese stock market has grown exponentially and become the second-largest in the world (Pan and Mishra, 2018). It plays an important role in global financial system and involves many institutional investors (Carpenter et al., 2021; Leippold et al., 2022). As a critical indicator of contagion risk, the connection among sectors has closely become intertwined (Zhang et al., 2020). The risk will affect other sectors once one sector encounters a risk shock, and the obvious linkage effect among sectors can trigger off risk contagion in the stock market (Zhang et al., 2020; Choi, 2022). The complete industrial chain of Chinese stock market helps the country quickly overcome the adverse effects. Zhou et al. (2023) analyzed the time-varying risk spillovers between sectors among industrial chain in the stock market. Xu et al. (2022) demonstrated that the upstream, midstream, and downstream sectors have varying impacts on the spread of risks under extreme market conditions. He et al. (2020) described the market performance and response trends in different sectors within the supply chain in response to the COVID-19 pandemic. Hence, it is essential to study the contagion risk among sectors and analyze how sectors at different positions within the industrial chain experience distinct changes.

In this paper, we propose a framework to specify and predict the contagion risk in Chinese stock market based on ADCC-GJR-GARCH and Chart GCN model. Firstly, financial networks are constructed to calculate correlation and centrality which are extracted to specify the contagion risk based on the ADCC-GJR-GARCH model. Then, technical charts are transformed into graphs through chart similarity, and each node in the graph is allocated technical indicators. Finally, the Chart GCN is used to extract the features in the graph and predict the contagion risk. In the experiment, this paper selects 28 sectors referring to Shenwan primary industry indices during 2007–2022 and applies a sliding time window to verify our proposed framework. Two conclusions are drawn as follows. First, we specify the contagion risk by testing its consistency with the occurrence of financial crisis. Second, our model proves the effectiveness of extracting technical chart features in contagion risk prediction based on the comparison with RNN, GRU, LSTM, CNN, MLP, and PATCHY-SAN.

Our research has made the following contributions. First, the feature information from both time series charts and technical indicators is fully exploited by the Chart GCN model to predict the contagion risk. Second, the contagion risk in the Chinese stock market can be specified as occurring when the correlation surges sharply and maintains a relatively high level over a period while three centrality indicators increase simultaneously. Hence, both the occurrence time of contagion risk and its duration can be obtained.

The remaining sections of this paper are organized as follows: Section 2 elaborates on the proposed methodology. In Section 3, the experimental setup, results analysis and robustness test are presented, followed by a conclusion in Section 4.

2. Research framework

In this section, we propose a contagion risk prediction framework that primarily comprises two components. Firstly, the contagion risk specification is determined by the correlation and centrality of financial network. Then, the Chart GCN extracts the features in the technical charts and technical indicators for predicting the contagion risk. Taking Chinese stock market as an example, the proposed framework is shown in Fig. 1.

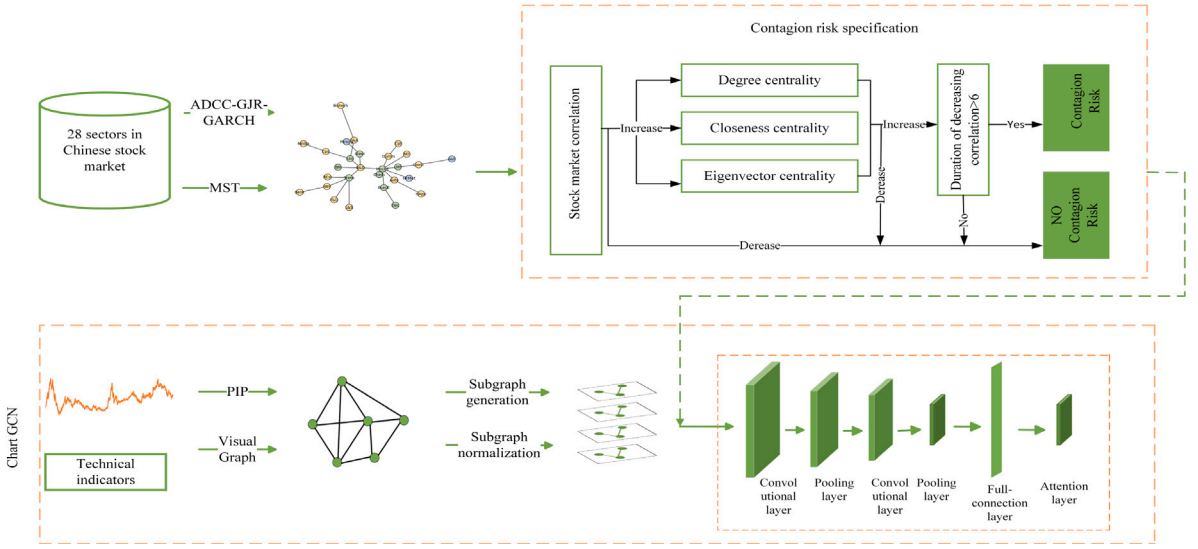


Fig. 1. The proposed framework of contagion risk prediction.

2.1. Contagion risk specification

2.1.1. Calculating correlation with the ADCC-GJR-GARCH model

Considering the potential asymmetric effects that often characterize financial series, we adopt the ADCC-GJR-GARCH model to calculate the correlation in the whole stock market network (Cappiello et al., 2006). The ADCC model is an extension of the dynamic conditional correlation (DCC) model which is capable of adaptively capturing market changes over time and better capturing asymmetry in financial market (Glosten et al., 1993). The GJR-GARCH model fits financial market data effectively and measures contagion risk accurately by considering leverage effects and asymmetry in volatility (Asif and Frömmel, 2022).

The ADCC-GJR-GARCH model mainly consists of two components, which involve establishing the ADCC model and building the GJR-GARCH model. The formulas for calculating the correlation coefficient in ADCC-GJR-GARCH model are as follows. For the single-variable GJR-GARCH variance equation of the ith asset return series $r_{i,t}$, the equation for its conditional variance $\sigma_{i,t}^2$ is:

$$\sigma_{i,t}^2 = \omega_i + \sum_{j=1}^{p}(\alpha_{i,j} + \gamma_{i,j} I_{t-j})\epsilon_{i,t-j}^2 + \sum_{k=1}^{q}\beta_{i,k}\sigma_{i,t-k}^2 \tag{1}$$

where $\omega_i > 0$, $\alpha_{i,j} \geq 0$, $\beta_{i,k} \geq 0$, and $\sum_{j=1}^{p}(\alpha_{i,j} + \gamma_{i,j}) + \sum_{k=1}^{q}\beta_{i,k} < 1$. $\epsilon_{i,t}$ is the residual of asset i at time t, that is $\epsilon_{i,t} = r_{i,t} - \mu_{i,t}$, and $\mu_{i,t}$ denotes the conditional mean prediction of $r_{i,t}$ based on a certain mean model such as the ARMA model. I_{t-j} is an indicator function, where $I_{t-j} = 1$ when $\epsilon_{i,t-j} < 0$, and $I_{t-j} = 0$ when $\epsilon_{i,t-j} \geq 0$. The standardized residual $z_{i,t}$ can be calculated by:

$$z_{i,t} = \frac{\epsilon_{i,t}}{\sqrt{\sigma_{i,t}^2}} \tag{2}$$

Suppose we have N asset return series, the elements $Q_{ij,t}$ $(i, j = 1, \ldots, N)$ of the dynamic conditional covariance matrix Q_t are calculated as follows:

$$Q_{ij,t} = (1 - \sum_{m=1}^{M}\alpha_m - \sum_{n=1}^{N}\beta_n)\overline{Q}_{ij} + \sum_{m=1}^{M}\alpha_m z_{i,t-m} z_{j,t-m} + \sum_{n=1}^{N}\beta_n Q_{ij,t-n} \tag{3}$$

where $\overline{Q}_{ij}$ is the long-term average covariance, and α_m and β_n are the model parameters. The dynamic conditional correlation coefficient ρ_{ij}^t is:

$$\rho_{ij}^t = \frac{Q_{ij,t}}{\sqrt{Q_{ii,t} Q_{jj,t}}} \tag{4}$$

where $Q_{ii,t}$ and $Q_{jj,t}$ are the diagonal elements of the dynamic conditional covariance matrix Q_t.

2.1.2. Extracting the centrality of complex networks

We represent each sector as a node and the distance derived from correlation transformation as edges to form a connected graph. The formula for converting the correlation between sectors into distance is as follows:

$$d_{ij}^t = \sqrt{2(1 - \rho_{ij}^t)} \tag{5}$$

where d_{ij}^{t} denotes the distance between variables i and j in time t, and ρ_{ij}^{t} represents the correlation between variables i and j in time t. In this process, the correlation coefficient matrix becomes the distance matrix that can be used to construct the financial network. To the best of our knowledge, the minimum spanning tree (MST), which could extract the most important connectivity information from the stock market while preserving the simplest network structure, has been widely utilized (Mantegna, 1999; Samitas et al., 2020). Considering that the weights of all edges carry practical significance in specifying contagion risk, we employ Kruskal's algorithm to construct the MST network to explain the interdependence between stock market sectors. Kruskal's algorithm, being independent of the network's specific shape or layout, is suitable for both sparse and complex topological networks. As long as the edge weights and node connectivity are determined, the MST construction can be effectively implemented. Thus, the financial network for the Chinese stock market is constructed, and subsequent changes in centrality indicators can be captured.

Three centrality indicators include degree centrality, eigenvector centrality, and closeness centrality are utilized to characterize the changes that occur in financial networks of stock markets (Samitas et al., 2020; He et al., 2022). Degree centrality and eigenvector centrality are crucial for identifying the most connected sectors and their influence within the network. Closeness centrality quantifies the efficiency of a sector in reaching other sectors. Their differences and calculation processes are as follows:

Degree centrality (D_i) quantifies the importance or influence of a node based on the number of connections it has with other nodes and the equation is:

$$D_i = \frac{k_{\cdot i}}{(n-1)} \tag{6}$$

where k_i is the degree of a sector which represents the sum of the elements in the i_{th} row of the matrix A, and n is the number of columns in the matrix.

Eigenvector centrality (E_i) assesses the importance or influence of a node based on the concept of eigenvectors and eigenvalues. It takes into account the number of connections of a node and the centrality of its neighbors:

$$E_i = \gamma \sum_{j=1}^{n} A_{ij} E_j \tag{7}$$

where γ is a constant and the largest eigenvalue of the adjacency matrix, $\sum_{j=1}^{n} A_{ij} E_j$ denotes the sum of the products of the j_{th} row elements of matrix A and the corresponding elements of vector E, and n is the number of columns in matrix A.

Closeness centrality (C_i) measures the degree of closeness or accessibility of a node to other nodes in the network. It quantifies how quickly or easily a node can reach other nodes in terms of shortest path distances:

$$C_i = \frac{1}{L_i} = \frac{N-1}{\sum_{(i,j)} d_{ij}} \tag{8}$$

where L_i is the average shortest path length from node i to all other nodes, N is the total number of nodes, and $\sum_{(i,j)} d_{ij}$ represents the sum of the shortest path lengths between all pairs of nodes.

2.1.3. Specifying the contagion risk

We specify the contagion risk in financial network of the Chinese stock market by using daily correlations and corresponding centrality indicators based on the following four observations. Firstly, in the event of a major crisis, a sharp increase in market correlation leads to increased contagion risk (Samitas et al., 2020; He et al., 2022; Xu et al., 2022). Secondly, an increase in correlation of financial network is often accompanied by an increase in centrality. Changes of stock market correlation would affect the change of network structure, the central node and centrality of the network (Li et al., 2021b; Xu et al., 2022). Thirdly, heightened correlations and centrality in the stock market are indicative of increased contagion risk (Samitas et al., 2020; He et al., 2022). The overall market correlation tends to remain at a relatively high level during crises (Liu et al., 2019, 2020). Finally, at the end of contagion risk, we consider a sustained period of decline in correlations between financial networks (Yang et al., 2014; He et al., 2022). The specific assumptions about how the network structure changes during contagion risk are illustrated in our proposed framework.

This dramatic increase in correlation is accompanied by an increase in the three centralities, which is regarded as the preliminary stages of contagion risk. During the occurrence of contagion risk, the correlation remains at a sustained and relatively high level. If the duration of decreasing correlation persists for more than six periods, we consider it to be the end of the contagion risk phase. Therefore, the contagion risk can be specified as occurring when the correlation surges sharply and maintains at a relatively high level over a period while three centrality indicators increase simultaneously. It can serve as the output variable Y for the Chart GCN model introduced in the following subsection, i.e.

$$Y = \begin{cases} 0 & \text{if no contagion risk} \\ 1 & \text{otherwise} \end{cases} \tag{9}$$

To address this binary classification problem, the technical indicators and technical charts of the Shanghai-Shenzhen 300 stock market are taken as the input variables of the prediction model.

2.2. Chart GCN model construction

We present the Chart GCN model to make better use of technical charts to predict contagion risk. Specifically, the model introduces a novel graph similarity method composed of perceptive importance points (PIP), visibility graphs (VG) and the GCN for prediction.

2.2.1. Converting technical charts

Our method consists of two main components that PIP calculates the importance score and VG transforms the node sequences into a graph.

PIP is employed to recognize technical charts, with the aim of calculating the importance scores for each node and identifying crucial nodes in the time series. Specifically, given a time series sample, the form of each node in the time series can be expressed as $\{(t_1, c_1), (t_2, c_2), (t_3, c_3), \ldots, (t_n, c_n)\}$. Here, t represents the time point and c represents the corresponding closing price. We obtain two lists containing important nodes and importance scores corresponding to important nodes using the PIP method. The important formula for calculating importance scores in PIP is:

$$d_{E(x_k, x_1, x_n)} = \sqrt{(t_k - t_1)^2 + (c_k - c_1)^2} + \sqrt{(t_n - t_k)^2 + (c_n - c_k)^2}. \tag{10}$$

The method of natural visual graphs establishes the relationship between time series and complex networks, enabling us to analyze time series from the perspective of graph theory. The natural visual graph method is utilized to transform the extracted key node time series into a graphical representation (Lacasa et al., 2008). Suppose $T = \{t_1, t_2, t_3, \ldots, t_n\}$ is the set of n data points of a discrete time series of significant nodes, two arbitrary data values (t_a, c_a) and (t_b, c_b) in the time series T $(a < b)$ have visibility at time points a and b if they satisfy the following visibility criterion:

$$\forall c \in (a, b), \quad c_c < c_a + \frac{c_b - c_a}{t_b - t_a}. \tag{11}$$

2.2.2. Generating and normalizing the subgraphs

Based on the above visual graph, we first generate new subgraphs and then normalize the receptive field of each node within these subgraphs. Firstly, we use the methods of node sequence selection and neighborhood assembly to generate subgraphs (Niepert et al., 2016; Li et al., 2022). In the node selection process for the subgraph, the first round of sorting is performed based on importance scores. In the case where two nodes have equal importance scores, the second round of sorting is conducted based on the degree of the nodes in the graph. If the degrees are also equal, the third round of sorting is then carried out based on the average importance scores of their neighboring nodes. Following the completion of the aforementioned sorting steps, the top N nodes are extracted to serve as the nodes of the subgraph. For the selected N subgraph nodes, a local neighborhood of size g needs to be constructed around each of them (Niepert et al., 2016; Li et al., 2022). To perform the selection of the node acceptance domain, we employ the breadth-first algorithm from graph traversal to satisfy the minimum subgraph size requirement of k. It is important to note that the value of k may not necessarily be equal to g.

After generating the subgraphs, we apply the principle of graph similarity to normalize the subgraphs. The process is analogous to the first-layer convolution operation of a CNN extracting local features from images (Niepert et al., 2016; Li et al., 2022). The entire process is roughly divided into three steps. Firstly, the nodes within the subgraph are taken as root nodes. Secondly, for each root node, breadth-first search is applied to filter nodes at each depth. Lastly, the nodes are sorted based on different depths. In the case of nodes at the same depth, sorting is done based on their importance scores. If the importance scores are equal, the degree of the nodes is then used for sorting. Following the aforementioned sorting steps, a maximum of k sorted neighbor nodes is obtained. If g is smaller than k, g nodes are extracted from the sorting as neighbor nodes. If g is larger than k, dummy nodes are added to ensure the count of neighbor nodes matches g. To enhance the efficiency of contagion risk prediction, each time point of a stock is augmented with f technical indicators. Consequently, the data is organized in a three-dimensional format $N \times g \times f$, where N denotes subgraphs comprising g nodes, and each node corresponds to f technical indicators.

2.2.3. Constructing the predictive model

Our prediction network structure employs the classic CNN architecture, named LeNet (Lecun and Bottou, 1998). Additionally, we integrate attention layers within the LeNet framework to enhance its performance. The input to the first convolutional layer is a three-dimensional feature $N \times g \times f$, and a convolutional kernel K_1 of size $g \times f$ is employed to extract node information. Next, the convolutional kernel K_2 is employed for the convolution on the N nodes in the graph. Therefore, adequate information has been extracted from both the node and subgraph dimensions of the graph. To enhance the model's capability for weight allocation to input data, we integrate attention layers into CNN. First, attention scores are calculated by taking the dot product of the input data and the attention weights:

$$Attention\ Scores = \text{inputs} \cdot attention_weights. \tag{12}$$

Then, the softmax function is applied to transform the attention scores into attention weights:

$$Attention\ Weights = \text{softmax}\ (Attention\ Scores). \tag{13}$$

Finally, we obtain the ultimate attention output by element-wise multiplication of the input data and the attention weights:

$$Attention\ Output = \text{inputs} \cdot Attention\ Weights. \tag{14}$$

Table 1
28 sectors in the Chinese stock market.

Name	Code	Abb.	Industrial chain
Agriculture and Forestry	801 010	A&F	
Mining	801 020	MIN	Upstream
Non-ferrous Metals	801 050	NFM	
Chemical	801 030	CHE	
Steel	801 040	STE	
Electronic	801 080	ELE	
Light Manufacturing	801 140	LMF	
Utilities	801 160	UTI	Midstream
Transportation	801 170	TRA	
Building Materials	801 710	BUM	
Electrical Equipment	801 730	EEQ	
Mechanical Equipment	801 890	MEQ	
Household Appliances	801 110	HAP	
Food and Drink	801 120	F&D	
Textile and Apparel	801 130	T&A	
Biotechnology	801 150	BIO	
Real Estate	801 180	RES	
Commercial Trade	801 200	COT	
Leisure and Services	801 210	L&S	
Composite	801 230	COS	Downstream
Building and Decoration	801 720	B&D	
National Defense	801 740	NAD	
Computer	801 750	COP	
Media	801 760	MED	
Communications	801 770	COU	
Bank	801 780	BAN	
Non-bank Financial	801 790	NBF	
Automobile	801 880	AUT	

3. Experimental results

3.1. Data

In this paper, the data of Chinese stock market is utilized for contagion risk specification and prediction. Our sample comprises 28 sectors in the Shenwan primary industry indices. The Shenwan primary industry indices categorize stocks in the A-share market into diverse sectors based on specific sector classification criteria. These sectors span multiple fields of economic activities, including but not limited to energy, finance, industry, and information technology. The Shenwan primary industry indices officially expanded from 28 to 31 sectors on July 30, 2021. Given that the period of our data ranges from 2007 to the end of 2022, we select the original 28 sectors for our study.[1] [2] By analyzing the different positions of these sectors in the industrial chain, it can be divided into upstream, midstream, and downstream (Xu et al., 2022). To be specific, the names, codes, abbreviations, and positions within the industrial chain of 28 sectors are listed in Table 1. In addition, the sample period is ranging from 5 January 2007 to 30 December 2022, which covers the global financial crisis, the 2015 Chinese stock market turbulence, the COVID-19, the Russia–Ukraine conflict and other periods. All data can be downloaded from the Choice and Wind databases.

For each series, the daily returns are computed using $100\times\log\left(\frac{p_t}{p_{t-1}}\right)$, where p_t and p_{t-1} are the daily closing prices at time t and $t-1$, respectively. Descriptive statistics of daily returns for the used variables are summarized in Table 2. We notice that the mean of returns of the Chinese stock market in the industrial chain are positive except steel. Another noteworthy statistics is the standard deviation of the National defense, which is the highest at 2.347. Meanwhile, the standard deviation of the Utilities returns is the lowest among all the sectors. Each of these return series shows high skewness and kurtosis values, which suggests that extremes are more common in Chinese stock market. In addition, for the unit root tests, we perform the Augmented Dickey–Fuller (ADF), which confirms that all the return series are stationary. Hence, we use these 28 sectors to build financial networks as well as set five days to observe changes in correlation and centrality (Samitas et al., 2020).

We collect the CSI-300 Index of the same period which represents the overall stock market situation to predict the contagion risk. The CSI-300 Index includes the most significant stocks in the Chinese A-share stock market. We assemble the closing price and calculate eight classes of trading indicators that include Turnover Rate, Price-to-Earnings Ratio (P/E Ratio), Trading Volume, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Moving Average (MA), Bollinger Bands (BOLL), and Stochastic Oscillator (KDJ) according to Lu et al. (2021) and Li et al. (2022). Table 3 gives the definition of all eight indicators.

[1] https://tushare.pro/document/2?doc_id=181
[2] https://www.swsresearch.com/institute_sw/allIndex/releasedIndex

Table 2
Descriptive statistics for 28 stock market sectors.

Sectors	Mean	Median	Std.	Max	Min	Kurtosis	Skewness	ADF
A&F	0.035	0.116	2.070	9.173	−9.763	2.631	−0.533	−24.96999***
CHE	0.023	0.158	1.857	8.320	−9.176	2.844	−0.726	−14.78879***
STE	−0.003	0.083	2.083	9.442	−10.297	2.792	−0.404	−13.46387***
NFM	0.022	0.114	2.333	9.235	−10.129	2.061	−0.418	−28.15395***
ELE	0.036	0.153	2.153	9.121	−10.245	2.103	−0.683	−34.51749***
HAP	0.054	0.089	1.933	9.074	−9.441	2.504	−0.358	−14.59573***
F&D	0.057	0.087	1.839	8.698	−9.315	2.271	−0.338	−23.70984***
T&A	0.012	0.138	1.929	8.934	−9.699	3.879	−0.868	−14.79389***
LMF	0.020	0.172	1.895	9.382	−10.018	3.773	−0.890	−14.20919***
BIO	0.051	0.129	1.850	8.866	−9.110	2.610	−0.578	−15.46944***
UTI	0.011	0.108	1.726	8.619	−9.812	4.012	−0.783	−12.87035***
TRA	0.004	0.080	1.797	9.496	−10.051	4.046	−0.698	−15.07345***
RES	0.013	0.022	2.102	9.404	−9.753	2.760	−0.451	−23.52114***
COT	0.011	0.120	1.856	9.275	−9.679	3.229	−0.715	−15.16362***
L&S	0.045	0.167	2.142	9.327	−9.798	2.482	−0.507	−21.49966***
COS	0.033	0.195	1.988	8.818	−9.741	2.876	−0.835	−14.56174***
BUM	0.032	0.111	2.086	9.030	−10.018	2.590	−0.521	−57.46275***
B&D	0.015	0.060	1.868	9.422	−0.492	3.720	−0.492	−13.49103***
EEQ	0.045	0.138	2.030	9.462	−9.158	2.185	−0.488	−12.37170***
NAD	0.027	0.142	2.347	9.016	−10.229	2.527	−0.438	−29.28176***
COP	0.038	0.116	2.182	9.455	−10.112	1.959	−0.510	−15.38948***
MED	0.008	0.100	2.128	9.533	−10.528	2.302	−0.562	−24.74264***
COU	0.010	0.098	2.044	9.347	−9.894	2.808	−0.452	−15.54084***
BAN	0.013	−0.051	1.806	9.550	−10.505	4.917	0.014	−10.23270***
NBF	0.010	−0.046	2.229	9.489	−10.197	3.085	−0.081	−11.19684***
AUT	0.031	0.109	2.035	8.960	−9.784	2.865	−0.578	−14.38545***
MEQ	0.025	0.119	1.943	9.338	−10.002	3.125	−0.691	−14.48484***
MIN	0.016	0.025	2.405	9.402	−10.271	2.161	−0.211	−15.28953***

Note: *, **, and *** represents significance at 10%, 5%, and 1% level respectively. The Augmented Dickey–Fuller (ADF) is used for stationarity testing.

Table 3
Definitions of the eight indicators.

Indicators	Definition
Turnover Rate	It is an important indicator for measuring the liquidity strength of stocks.
Price-to-Earnings Ratio (P/E Ratio)	It is one of the key indicators for evaluating the stock valuations.
Trading Volume	It refers to the quantity of a certain stock traded on the stock market within a specific unit of time.
Relative Strength Index (RSI)	It is used to measure the speed and magnitude of stock price changes.
Moving Average Convergence Divergence (MACD)	One of its most important applications is to judge the trend of stock prices.
Moving Average (MA)	It is a method of smoothing out price fluctuations by calculating the average value of stock prices over a certain time period.
Bollinger Bands (BOLL)	It helps to judge the fluctuation range of stock prices and the signals of trend changes.
Stochastic Oscillator (KDJ)	It is mainly used to measure the position of stock prices within the fluctuation range during a certain period.

In terms of data setting, we can obtain training, validation, and test sets by partitioning the time series sequentially. In experiment, we split the data from 2007 to 2022, with the first 90% designated as the training set and the remaining 10% as the test set. Within the training set, 80% of the data is used for training, and the remaining 20% is designated as the validation set. In the empirical analysis, the ratio of positive to negative sample is approximately 1 : 6, making it an imbalanced dataset for prediction. Hence, we employ the method of weight adjustment to handle the imbalanced data situation (Tao et al., 2019).

3.2. Contagion risk specification result analysis

3.2.1. Explanation of correlation and centrality

By calculating the correlation between different sectors in industrial chain, Fig. 2 illustrates the dynamic changes in overall correlation within the Chinese stock market, as well as those within the upstream, midstream, and downstream. The sudden rise in correlations within the stock market industry is associated with the turbulence of stock market and financial crises. Most striking evidence is that sectors at different positions within the industrial chain exhibit almost identical behavior, especially in the upstream and midstream. This phenomenon shows that even though different sectors in the stock market belong to different positions in the

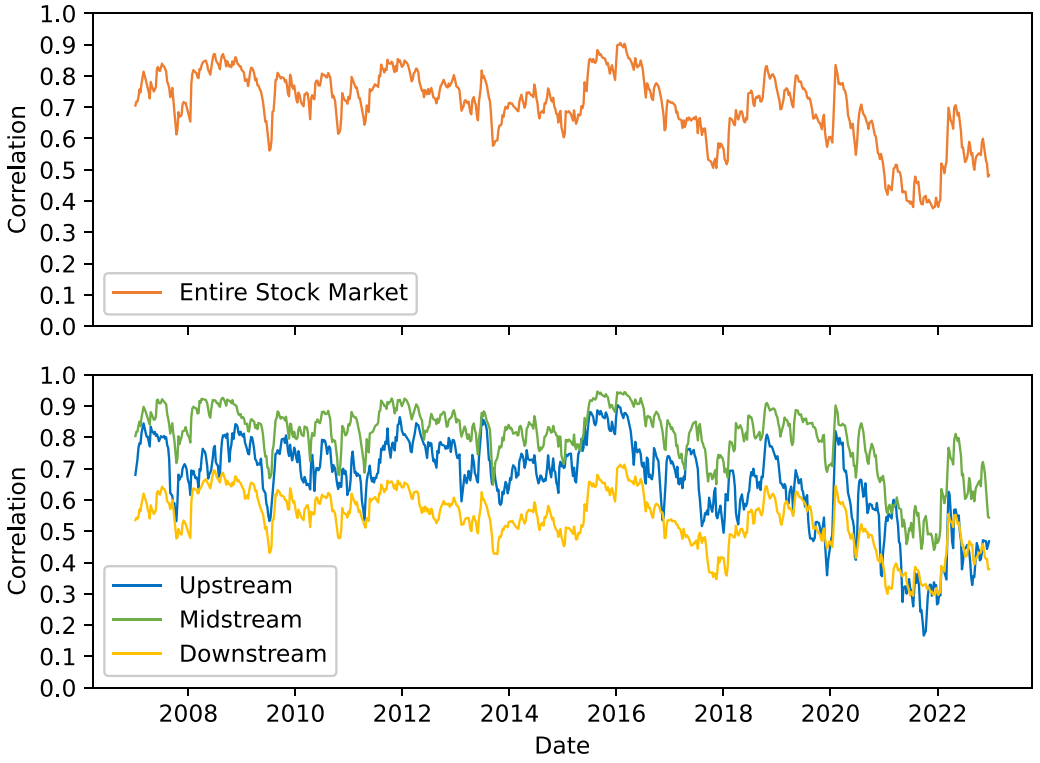


Fig. 2. Correlation of the Chinese stock market based on ADCC-GJR-GARCH.

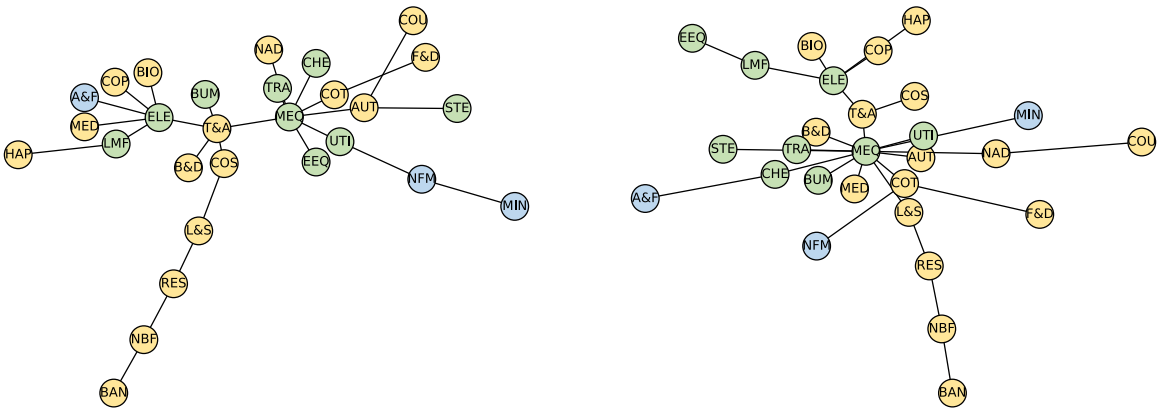


Fig. 3. Complex financial networks before and after the 2008 crisis.

industrial chain, the dynamics seem to show similar conditional correlation behavior. This behavior remains consistent throughout the entirety of our dataset.

Through the financial network of the global financial crisis and COVID-19 constructed by the MST in Figs. 3 and 4, we can observe changes in the structural information of the financial network topology. Specifically, each financial network extracted from the average correlation of the entire sample reveals close connections between sectors, with upstream and midstream sectors dominating the spread of contagion risk. The topology of the financial network evolves over time, which indicates that the position of various sectors in the network structure changes over time and also reflects the impact on the stock market in terms of contagion risk. In financial crises and epidemic events, similar characteristics exist between stock markets in different situations. This indicates that the financial network topological features of the stock market can explain the occurrence and spread of contagion risk, and possess important explanatory power during major unforeseen events.

The dynamic evolution of the centrality of the highest centrality sector over time can be observed in Figs. 5 to 7, which are constructed using the data on the centrality of the financial network structure. Following the occurrence of contagion risk, three

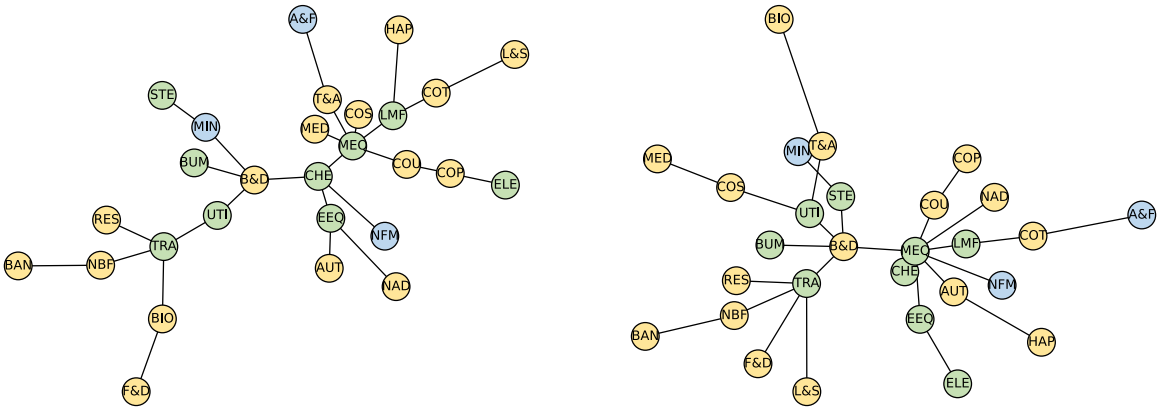


Fig. 4. Complex financial networks before and after the COVID-19.

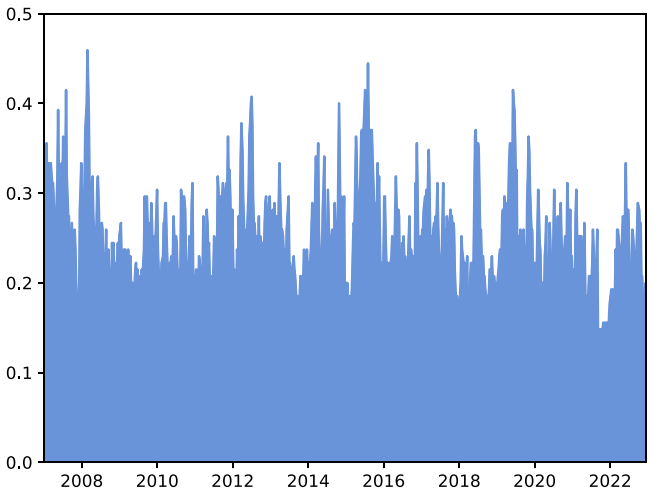


Fig. 5. Changes in the highest degree centrality in financial networks.

centralities in the financial network experienced varying degrees of increase. The increased interdependence within the Chinese stock market network suggests that changes may partly stem from the effect of contagion risk. The fluctuations in correlations largely follow the fluctuations in centrality. Specifically, significant shocks in correlations trigger substantial fluctuations in centrality. With the increasing impact of correlation shock, the amplitude of centrality fluctuation also increases.

3.2.2. Specification results

Comparing the correlation and centrality of the dynamic financial network with the actual crisis dates, Fig. 8 confirms the accuracy of the contagion risk specification. Firstly, the results affirm the presence of contagion risk on dates where a significant increase in correlation and centralities is observed. Specifically, in most instances where we observe a substantial increase in correlation, we have identified an increased likelihood of contagion risk. Secondly, we find that during the period of contagion risk, correlations consistently remain at a relatively high level. Finally, correlations have continued to decline more than six periods as the contagion risk has ended. Our contagion risk specification accurately identifies the occurrence of most contagion risk compared with the actual crisis dates.

Taking financial and economic crisis as an illustrative case, the 2008 U.S. subprime mortgage crisis exerted a profound and far-reaching influence on the Chinese stock market. During this crisis, a substantial surge in the correlation among sectors within the financial network was detected by calculating and monitoring the correlation and centrality of sectors. In comparison to the pre-crisis period, the average correlation escalated to 0.78 and persisted at a relatively elevated level over an extended period, maintaining a highly correlated state for 300 trading days. After the crisis, the overall correlation and centrality among sectors in the Chinese stock market gradually showed a trend of a continuously declining curve. Similarly, the 2010 Eurozone debt crisis also had a significant and conspicuous impact on the Chinese stock market. The correlation of sectors within the financial network experienced a remarkable augmentation. Relative to the pre-crisis period, it ascended to 0.74 and remained at a relatively high level for a prolonged duration, being highly correlated for 85 trading days before gradually tapering off. In 2013, owing to the liquidity

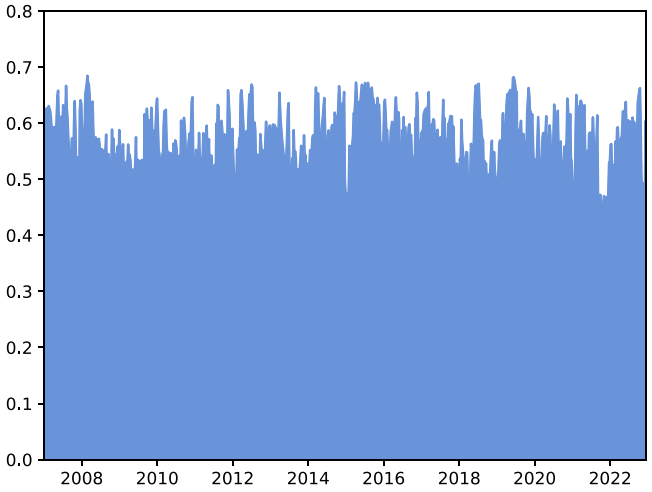


Fig. 6. Changes in the highest eigenvector centrality in financial networks.

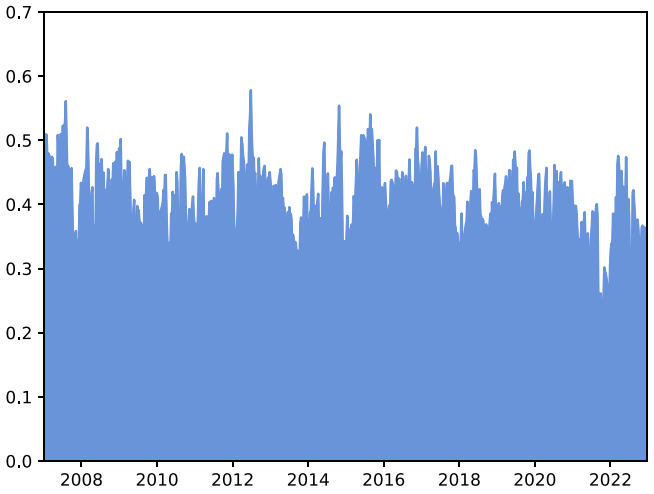


Fig. 7. Changes in the highest closeness centrality in financial networks.

crunch, the network correlation of the Chinese stock market witnessed a sharp spike, with the average correlation reaching 0.76 and remaining highly correlated for approximately 41 trading days. In 2015, the economic crisis precipitated a precipitous decline, accompanied by a soaring of the internal network correlation and centrality, where the average correlation peaked at 0.85 and remained highly correlated for around 265 trading days.

Taking external shock crisis as an illustrative case, the 2020 global public health crisis swept across the world, leading to widespread panic. This non-financial crisis also had an effect on the financial network of the Chinese stock market. Through analysis, the average correlation rose to 0.83 and remained at a relatively high level for about 72 trading days. In 2022, the geopolitical crisis prompted the financial network of the Chinese stock market to respond quickly. The average correlation rose to 0.69 and remained highly correlated for about 71 trading days. These external shock crises fully validate the consistency between the contagion risk in the stock market and actual crises, highlighting the close relationship between external events and the internal structure of the market network.

3.3. Comparative analysis of prediction model

We first elaborate on the settings utilized to evaluate the models, which include parameters of the neural networks, evaluation metrics and baseline models. Secondly, the evaluation results are reported.

3.3.1. Parameter setting

This paper aims to introduce the effectiveness of utilizing the Chart GCN algorithm to predict the data of Chinese stock market. We compare the performance of the proposed method with that of the neural network models applied in the following experiment.

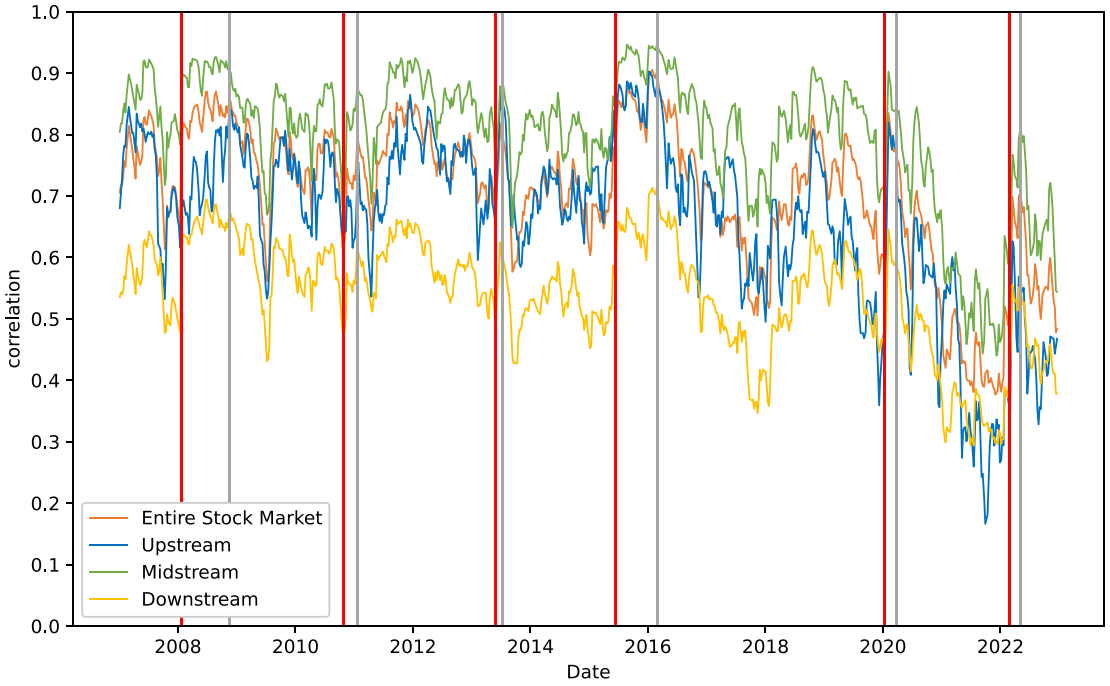


Fig. 8. Contagion risk specification results. Notes: the red line represents the onset of contagion risk and the gray line signifies the end of contagion risk. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Parameters setting of prediction models.

Model	RNN LSTM MLP GRU	CNN	PATCHY-SAN Chart GCN
settings	$units \in [32, 64, 128, 256]$	$filters \in [32, 64, 128, 256]$	$filters \in [32, 64, 128, 256]$
	$dropout \in [0.1, 0.2, 0.3]$	$dropout \in [0.1, 0.2, 0.3]$	$dropout \in [0.1, 0.2, 0.3]$
		$dense \in [32, 64, 128]$	$dense \in [32, 64, 128]$
		$pool_size \in [(1,1),(2,2),(3,3),(5,1),(5,2),(5,3)]$	$N \in [5, 10, 15, 20, 25]$
		$kernel_size \in [(1,1),(2,2),(3,3),(5,1),(5,2),(5,3)]$	$G \in [3, 4, 5, 6]$

These models used as baselines are MLP, RNN, LSTM, GRU, CNN, and PATCHY-SAN. In addition, the performance of each forecasting model is measured, regarding the binary classification problem of predicting whether there will be a contagion risk. If there is a contagion risk, the label is set to 1, otherwise, the label is set to 0 (Li et al., 2022). Referring to Weng et al. (2017), Hoseinzade and Haratizadeh (2019), and Li et al. (2020a), four evaluation metrics are selected to evaluate each model's performance, including recall (Recall), accuracy (ACC), F1-score (F1), and area under the curve (AUC). Accuracy calculates the ratio of the number of positive class samples to the total number of samples. A higher value indicates better classification performance. However, it may be partial toward the forecasting models that are prone to predict the more majority class. In an imbalanced category distribution, F1-score and AUC are more suitable to comprehensively capture the performance of models in full. Recall measures the model's ability to successfully identify positive classes. A larger recall value reflects a greater tendency to correctly predict contagion risk.

In the experiment, we employ a grid search approach to seek the parameters for all seven models. The settings for various types of model parameters are presented in Table 4. In all the baseline models mentioned above, the same common parameter settings are used. For example, the class_weight $\in \{0 : 1, 1 : 3.5\}$, the batch size is 64 and the time windows n of all indicators within {5, 10, 20, 30, 40, 50, 60, 100}. Specifically, for Chart GCN and PATCHY-SAN, the width of the top nodes N equals the average number of nodes within $\{5, 10, 15, 20, 25\}$, and the minimum subgraph size G within $\{3, 4, 5, 6\}$.

3.3.2. Prediction results

The comprehensive results of each model for our four metrics are presented in Table 5, where bold numbers represent the best performance among different evaluation metrics respectively. As it is shown, the proposed model demonstrates exceptional performance, achieving state-of-the-art results with a high AUC value, remarkable Recall and F1 score, as well as a highly competitive ACC. Specifically, our model demonstrates superior capability in capturing contagion risks compared to the baseline models, achieving a recall rate of 77.39% for its best performance model. The proposed Chart GCN exhibits the highest AUC value of 79.48%, surpassing the second-best baseline model by 1.32%. Moreover, it achieves the highest F1 score of 49.30%, outperforming all other baseline models. With that being said, the overall results (based on the AUC, ACC, F1, and Recall) prove that the proposed

Table 5
Prediction results of Chart GCN and baseline models.

	Recall(%)	ACC(%)	F1(%)	AUC(%)
LSTM	51.30 ± 13.65	83.40 ± 2.84	41.85 ± 3.06	69.51 ± 4.53
GRU	55.65 ± 8.43	**84.94 ± 1.89**	46.6 ± 1.15	72.27 ± 2.63
RNN	55.65 ± 22.94	84.48 ± 2.07	43.88 ± 11.46	72.00 ± 9.30
MLP	63.91 ± 13.58	82.78 ± 1.55	46.32 ± 3.43	74.61 ± 5.01
CNN	75.65 ± 4.84	80.36 ± 1.49	47.76 ± 2.48	78.32 ± 2.25
PATCHY-SAN	75.65 ± 11.20	79.27 ± 4.01	46.54 ± 2.51	77.70 ± 3.30
Chart GCN	**77.39 ± 5.92**	81.08 ± 1.96	**49.30 ± 0.99**	**79.48 ± 1.56**



Fig. 9. Contagion risk specification results in Hong Kong stock market. Notes: the red line represents the onset of contagion risk and the gray line signifies the end of contagion risk. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Chart GCN model is a good predictor for contagion risk. It is significant and justifies the effort needed to extract time series charts as features. And, the Chart GCN demonstrates superiority in graph feature extraction and risk prediction. In addition, the PATCHY-SAN model also leverages technical chart data for predictions, achieving a Recall of 75% and an AUC of 78% in the evaluation metrics. This indicates that the node ordering method PIP and convolutional neural network structure of Chart GCN enhance the performance of contagion risk prediction.

3.4. Robustness test

Hong Kong stock market that includes 13 sectors in the Hang Seng Composite Industry Index is utilized for robustness test to verify the effectiveness of the proposed framework. The empirical specification and prediction results are shown in Fig. 9 and Table 6. Two main results are summarized as follows:

Five occurrence times and their corresponding periods of contagion risk are specified, which are consistent with the actual crises. For instance, the financial stock crash in 2015 led to a sharp decline in the Hong Kong stock market, accompanied by a soaring of the internal network correlation and centrality, where the average correlation peaked at 0.74 and remained highly correlated for around 183 trading days. The COVID-19 pandemic, which caused the spread of panic, also had an impact on the stock market. The average correlation rose to 0.71 and remained at a relatively high level for approximately 172 trading days. However, this public health crisis did not completely come to an end. Instead, it once again led to a rapid increase in the average correlation of the financial crisis in the Hong Kong stock market in 2021 and remained at a relatively high level for about 100 trading days.

The Chart GCN model outperforms six other baseline models in terms of selected evaluation metrics. To be specific, the prediction results of the Chart GCN model possess the highest values of Recall, ACC, F1, and AUC, which are larger than those obtained by

Table 6
Prediction results in Hong Kong stock market.

	Recall(%)	ACC(%)	F1(%)	AUC(%)
RNN	43.89 ± 22.55	58.28 ± 2.32	39.84 ± 14.02	55.11 ± 5.07
MLP	65.10 ± 8.43	60.20 ± 0.88	53.92 ± 2.76	61.33 ± 1.82
LSTM	72.38 ± 7.26	60.12 ± 2.00	56.43 ± 3.73	62.82 ± 3.16
PATCHY-SAN	71.15 ± 8.75	61.33 ± 1.13	56.68 ± 3.42	63.49 ± 2.41
CNN	77.69 ± 10.11	61.36 ± 4.26	59.02 ± 5.78	65.13 ± 5.50
GRU	75.75 ± 9.72	61.65 ± 3.02	58.41 ± 5.07	64.75 ± 4.48
Chart GCN	**81.41 ± 9.04**	**63.11 ± 2.75**	**61.11 ± 4.46**	**67.14 ± 4.11**

RNN, MLP, LSTM, PATCHY-SAN, CNN, and GRU. These results prove that our proposed framework possesses stronger robustness across different stock markets.

4. Conclusion

This paper proposes a framework for contagion risk prediction by comprehensively considering the features of technical charts and technical indicators based on the Chart GCN model. The framework comprises two main components. The first component involves using correlation and centrality of the financial network to specify contagion risk. The second one involves employing Chart GCN to extract features from the time series chart and predicting contagion risk.

Taking the 28 sectors in the Chinese stock market as an example, two main results are obtained. First, by referring to the actual situation, we find that the contagion risk specification results determined by the correlation and centrality exhibit higher accuracy. The financial network structure indicates a sharp increase in network correlations during crises. The correlations have continued to decline as the contagion risk has ended. These correlations influence changes in centrality, with all three centrality measures simultaneously increasing. The contagion risk within the network is transmitted from upstream to downstream sectors. Second, the proposed model proves the effectiveness of extracting technical chart features and the superior performance of GCN in handling graph data. In terms of the empirical results of the prediction model, we compare it with LSTM, GRU, RNN, MLP, CNN, and PATCHY-SAN. We find that Chart GCN exhibits overall better performance across different evaluation metrics. Particularly, the significant advantage in recall indicates its greater sensitivity in capturing the presence of contagion risk. In addition, the robustness of the proposed framework is verified in Hong Kong stock market.

In general, the contagion risk prediction model is applicable to market shocks that occur relatively quickly and whose impacts manifest rapidly in the short term. For example, the 2008 U.S. subprime mortgage crisis prompted the Chinese stock market financial network to react rapidly, with the correlation and centrality indicators rising simultaneously. The period of the contagion risk derived from it coincides with that of the stock market. However, the external events that maintain a long-term and gradual process have not been detected, such as the Sino-US trade friction incident. China and the United States have conducted many trade consultations and there has been a period of eased trade friction starting from March 9, 2018 (Li et al., 2020b).

Consequently, our research can be applied to contagion risk management for supervisors, policymakers and investors. For supervisors and policymakers, we have identified and predicted the contagion risk that poses the threat to the stability of the financial network. They could concentrate on monitoring the core sectors of the entire financial system. For investors, our search can help them accurately evaluate contagion risk and optimize portfolio allocations or investment decision-making. However, this paper still has several limitations, and future work can be expanded in the following aspects. First, it would be interesting to take into account the risk spillover in network construction. Second, the prediction models we plan to further pursue are graph neural network and graph attention network, which extract graph data and time series data more effectively.

CRediT authorship contribution statement

Zhensong Chen: Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **Wenjun Zhang:** Methodology, Formal analysis, Data curation, Software, Visualization, Writing – original draft. **Yinhong Yao:** Validation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data can be downloaded from the Choice and Wind databases. You can also find the data and code with the Mendeley dataset link: https://data.mendeley.com/datasets/6xy9d4bp28/1.

Data and code for: Contagion risk prediction with Chart Graph Convolutional Network: Evidence from Chinese stock market (Original data)

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